NALBUPHINE

Ceiling effect on respiration

Can precipitate withdrawal

CHEMICAL NAME (5α,6α)-17-(cyclobutylmethyl)-4,5-epoxymorphinan-3,6,14-triol (as hydrochloride)

CLASS Mixed opioid agonist–antagonist (κ agonist, partial μ antagonist)

TRADE NAMES Nubain · Nalbuphine HCl injection

THERAPEUTIC USES / INDICATIONS

- Moderate to severe pain (including obstetric labour analgesia).
- Pre- and post-operative analgesia; supplement to anaesthesia.
- Opioid-induced pruritus (small doses).

ADMINISTRATION

- **IV** slowly over 2–3 min, **IM** or **SC**.
- Not available orally.
- Usually not a controlled drug in many countries (lower abuse potential).

DOSAGES

- **Adult:** 10 mg per 70 kg SC/IM/IV every 3–6 h; max 20 mg single dose, 160 mg/day.
- **Children:** safety not established in US label; use only per specialist/local protocol.
- **Opioid-induced pruritus:** low doses used off-label (per protocol).
- Reduce in hepatic or renal impairment.

PHARMACOKINETICS

- **Onset:** IV 2–3 min; IM/SC <15 min. **Peak:** IV 30 min. **Duration:** 3–6 h.
- Metabolised in liver.
- Excreted in faeces and urine. **t½:** about 5 h.

PHARMACODYNAMICS

- **κ-opioid agonist** → analgesia and sedation.
- **Partial μ antagonist** → ceiling effect on respiratory depression; can reverse μ effects of other opioids.
- Less euphoria, constipation and dependence than morphine.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Sedation (common), dizziness.
- Sweating, nausea, dry mouth.
- Headache.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression (limited by ceiling but still possible).
- Withdrawal symptoms in opioid-dependent patients.
- Fetal bradycardia when used in labour.
- Dysphoria, hallucinations (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to nalbuphine.
- **Caution:** opioid-dependent patients, head injury, respiratory disease, hepatic or renal impairment.

KEY INTERACTIONS

- **Pure μ agonists (morphine, fentanyl):** nalbuphine reduces their analgesia.
- **CNS depressants:** additive sedation.

NURSING MANAGEMENT

- Ask about current opioid use or dependence.
- Monitor RR, sedation, BP and pain response.
- Labour: monitor fetal heart rate; inform neonatal team.
- Keep naloxone available.
- Fall precautions due to sedation.

PATIENT EDUCATION

- You may feel sleepy; stay in bed or ask for help.
- Report breathing difficulty or unusual thoughts.
- Avoid alcohol and sedatives.

KEY POINTS

- κ agonist / μ antagonist.
- Can precipitate opioid withdrawal.
- Ceiling on respiratory depression.
- Used in labour and for opioid itch.

SOURCES CHECKED **[49]** Nalbuphine Hydrochloride, IM, IV, SC (US label) **[50]** Nalbuphine Hydrochloride, IM, IV, SC (US label, Hospira)

Full references p. 432–457